

Feiyang Wu

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Education

- 2025 – 2027 📖 **M.E. in Computational Science and Engineering**, Harvard University, Cambridge, MA.
Expected May 2027.
- 2021 – 2025 📖 **B.S. in Computer Engineering**, University of Illinois Urbana-Champaign, Champaign, IL.
Graduated with High Honors. GPA 3.97/4.00.
- 📖 **B.Eng. in Electronic and Computer Engineering**, Zhejiang University, Haining, China.
Dual degree through the ZJU-UIUC Institute. Class rank 1/64.

Publications

- 1 H. Chen, **F. Wu**, Y. Ma, M. Mete, P. Ye, J. Shen, C. Zhu, A. Ruggeri, K. Cheung, J. Mao, E. Adelson, J. Wu, R. D. Howe, and Y. Du, *Robot learning with visual predicted force*, Under review, ICRA 2027; co-first author, 2026. 🔗 URL: <https://visual-force.github.io/>
- 2 H. Chen, Y. Ma, S. Tian, X. Han, W. Huang, **F. Wu**, Y. Li, J. Wu, E. H. Adelson, and Y. Du, “CoStream: Composing simple behaviors for generalizable complex manipulation,” 2026. arXiv: 2606.26423 [cs.R0]. 🔗 URL: <https://arxiv.org/abs/2606.26423>
- 3 E. Song, W. Chai, G. Wang, Y. Zhang, H. Zhou, **F. Wu**, H. Chi, X. Guo, T. Ye, Y. Zhang, Y. Lu, J.-N. Hwang, and G. Wang, “MovieChat: From dense token to sparse memory for long video understanding,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 18 221–18 232. 🔗 DOI: 10.48550/arXiv.2307.16449

MovieChat has been cited 700+ times as of September 2026.

Research Experience

- 2024 – 2025 📖 **Project Leader**, *Diffusion and Flow Matching Policy with Torque Modality*.
- Built an **ARX** robotic-arm teleoperation data collection, training and evaluation pipeline to replicate results from Diffusion Policy (DP) and **Flow Matching** on basic and contact-demanding wiping and flipping tasks.
 - Incorporated joint **torque modality** into raw DP, observing improved action performance on contact-rich tasks.
- 2023 📖 **Contributor**, *MovieChat-1K Benchmark*.
- Co-authored the **CVPR 2024** paper **MovieChat** (700+ citations), a widely adopted benchmark for multimodal LLMs and vision-language models in long-form video understanding requiring video memory.
 - Built the data collection software stack and annotated a high-quality benchmark with Q&A pairs and 100-word summaries for 1000+ ten-thousand-frame videos, released on the Hugging Face Hub.

Industry Experience

- Jun – Aug 2026  **Software Development Engineer Intern**, Amazon, Boston, MA.
- Designed and built a computer-vision-based testing pipeline in **Rust** for Amazon's next-generation **dashcart**, connecting edge devices to AWS IoT Core over MQTT, using gRPC for inter-module communication, and resolving credential provisioning for newly onboarded cart units via AWS Lambda.
 - Built and deployed an automated agentic testing workflow on **aws AgentCore** that reads the cart's on-screen display to flag anomalies during rollout and streams results to a real-time monitoring pipeline, with the goal of removing the need for on-site engineers during beta provisioning on the test fleet.
- Jul – Aug 2025  **Machine Learning Engineer Intern (Manipulation)**, LimX Dynamics, Beijing, China.
- Contributed to the pre-release **FluxVLA Engine**, an end-to-end VLA engineering platform spanning data processing, fine-tuning, evaluation and real-robot deployment across 7 mainstream VLA models (OpenVLA, GROOT, $\pi_0/\pi_{0.5}$ and more) on Franka, UR3 and ALOHA hardware; now open-sourced (500+ stars, 60+ forks) and deployed on **Alibaba Cloud** to serve VLA training and inference.
 - Built a unified codebase for several VLA models including OpenVLA and π_0 , resolving environment inconsistencies between them and transplanting **OpenVLA-OFT** onto the **LIBERO** benchmark.

Teaching Experience

- Fall 2024  **Teaching Assistant**, CS 101: Introduction to Computing for Engineering & Science, ZJU-UIUC Institute.
- Taught Python-based computational problem solving; ran weekly labs and discussion sections on programming exercises, debugging and algorithm design.
 - Mentored students with little or no prior programming background through office hours, applying programming to data analysis, modelling and simulation.
- Spring 2023  **Teaching Assistant**, ECE 120: Introduction to Computing, ZJU-UIUC Institute.
- Led weekly discussion sections on Boolean algebra, sequential logic and finite state machines, reinforcing theory with applied problem solving.
 - Supervised laboratory sessions on circuit simulation, C programming and system-level design, covering hardware–software integration and LC-3 assembly.

Leadership & Awards

Leadership

- 2022 – 2025  **Team Captain**, RoboMaster Meta Robotics Team, ZJU-UIUC Institute.
- Captain across two terms (2022–2023 and 2024–2025), leading ~20 students across mechanical, computer vision and embedded subteams to design, build and debug competition robots.
 - Built the robot control system architecture and implemented computer-vision-based object detection and tracking used throughout the competition season.
 - Deployed **ros2-control** on NVIDIA Jetson Nano and ChibiOS (RTOS) on STM32 for precise motor and gimbal control; led a CUDA- and TensorRT-accelerated YOLO pipeline for real-time perception.

Honors and Awards

- 2025  **B.S. with High Honors**, University of Illinois Urbana-Champaign.
- 2023 – 2024  **Dean's List**, Grainger College of Engineering, UIUC.

Leadership & Awards (continued)

2023	■ Second Prize , RoboMaster University League, Shanghai.
2022 & 2023	■ National Scholarship of China.
	■ First Prize Scholarship , Zhejiang University.
	■ Second Prize Scholarship , ZJU-UIUC Institute.
2022	■ National Gold Prize , 'Internet+' Innovation and Entrepreneurship Competition.

Skills

Programming	■ C, Python, C++, Rust, MySQL, CUDA, x86 assembly, Verilog, JavaScript/Node.js, HTML/CSS.
Frameworks	■ PyTorch, TensorFlow, JAX, ROS2, TensorRT, MuJoCo, YOLO, OpenCV, ComfyUI, AWS.
Research Areas	■ LLMs, vision-language-action models, world models, generative models, reinforcement learning, control theory, embedded systems.
Languages	■ Mandarin Chinese (native), English (fluent).